

Model Analysis Report

Risk Assessment Model

MKM-RA-001

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Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Combined risk score by flood risk level and LTV ratio (no age/construction adjustment).

Flood Risk	LTV=50%	LTV=70%	LTV=85%	LTV=96%
Very Low	1.00	1.20	1.50	2.00
Low	2.00	2.40	3.00	4.00
Medium	4.00	4.80	6.00	8.00
High	7.00	8.40	10.0	10.0
Very High	9.00	10.0	10.0	10.0

Table 2: Combined risk score by construction type and age (LTV=85%).

Configuration	Low	Medium	High
Brick, 30yr	3.00	6.00	10.0
Brick, 80yr	3.30	6.60	10.0
Brick, 120yr	3.90	7.80	10.0
Timber, 30yr	3.90	7.80	10.0
Timber, 80yr	4.30	8.60	10.0
Steel, 30yr	2.40	4.80	8.40
Concrete, 30yr	2.70	5.40	9.50

Table 3: Value at risk (£) by risk level and flood depth (property value = £300,000).

Depth	Low	Medium	High	Very High
None	15,000.0	45,000.0	105,000.0	180,000.0
0.3m	30,000.0	45,000.0	105,000.0	180,000.0
0.5m	30,000.0	45,000.0	105,000.0	180,000.0
1.0m	75,000.0	75,000.0	105,000.0	180,000.0
2.0m	150,000.0	150,000.0	150,000.0	180,000.0
3.0m	240,000.0	240,000.0	240,000.0	240,000.0

Table 4: Insurance premium factor (fraction of property value) by risk level and flood depth.

Risk Level	d=0m	d=0.5m	d=1.0m	d=2.0m
Very Low	0.0010	0.0013	0.0015	0.0020
Low	0.0020	0.0025	0.0030	0.0040
Medium	0.0050	0.0063	0.0075	0.0100
High	0.0150	0.0187	0.0225	0.0300
Very High	0.0350	0.0438	0.0525	0.0700

Table 5: Distance risk decay factor.

Distance (km)	Risk Factor
0.0000	1.00
0.1000	1.00
0.5000	0.8000
1.00	0.6000
2.00	0.4000
5.00	0.2000
10.0	0.1000

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 6: Stress testing scenarios for Risk Assessment Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2 σ inputs	Medium	Pending
Severe stress	+3 σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 7: Performance metrics for Risk Assessment Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.